

Scientific Multimodal Cross-Attention Alignment & Neural Inversion Architecture

1. Heterogeneous Modalities

Multispectral / SAR Imagery

Optical & Radar Satellite Cubes
Sentinel-1/2, Landsat, MODIS
Irregular revisit & multi-scale
(X_{sat} in $\mathbb{R}^{T \times C \times H \times W}$)

Gridded Atmospheric Tensors

Global Gridded Reanalysis
ERA5, IFS Analysis, GFS
Uniform lat-lon / spherical grid
(X_{grid} in $\mathbb{R}^{B \times L \times H \times W}$)

Point Series & Waveforms

Sparse In-Situ Sensor Networks
Streamflow gauges, Seismograms
Weather stations, Ocean buoys
(X_{point} in $\mathbb{R}^{N \times T \times D}$)

Asynchronous Sensor Tracks

Asynchronous Soundings & Tracks
Radiosondes, Aircraft AMDAR
Satellite Altimetry (SSH Tracks)
(y_k at irregular t_k)

2. Cross-Modal Alignment Strategies

Strategy A: Early Heterogeneous Patching

Modality-Specific Patch Embeddings
Positional Harmonization: $PE(x, y, z, t)$
Continuous coordinate encodings
(SatMAE, Presto, Galileo)

Strategy B: Latent Cross-Attention Bottleneck

Fixed Learnable Latent Query Bank: Z in $\mathbb{R}^{M \times D}$
 $CrossAttn(Z, X_{\text{modality}}, X_{\text{modality}})$
Decouples irregular sensor density from backbone compute
Enables arbitrary multi-sensor conditioning
(Aurora, SkySense, ClimaX, Prithvi)

Strategy C: Dynamic Wavelength Hypernetworks

Continuous Spectral Wavelength Embedding: e_{lambda}
Dynamic Hypernetwork: $H_{\text{phi}}(e_{\text{lambda}}) \rightarrow W_{\text{kernel}}$
Zero-shot sensor adaptation across arbitrary bands
Continuous band interpolation (400nm - 2200nm)
(DOFA, AnySat)

3. Foundation Backbone & Inversion

Unified Spatio-Temporal Backbone

Spherical Harmonic Transforms (SFNO)
Multi-Mesh Message Passing (GraphCast)
3D Earth Transformers (Pangu, FuXi)
Score-Based Diffusion SDE (GenCast, SEEDS)
Preserves Kolmogorov Turbulent Spectra

Neural Variational Data Assimilation

Neural 4D-Var Unrolled Solvers
 $x_0^{(i+1)} = x_0^{(i)} - \Gamma_{\theta}(\text{grad } J)$
(4DVarNet, FengWu-4DVar)

Score-Based Posterior Sampling
 $dx = [f(x, t) - g^2(\text{grad log } p + \text{grad log } p(y|x))] dt$
(DiffDA, Score-based DA)
Sparse Observation $\rightarrow$ Gridded Initial State

Aligned Tokens

Latent Queries Z

Optimal x_0

Dynamic Kernels